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A Multimodal-based Framework for Smart ECG Report Generation

Tri-modal ECG signal, Computer Vision, and LLM for Automatic ECG Report Generation

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Paper, refs & contact

1. Motivation

Clinical Problem. Electrocardiograms (ECGs) are essential for cardiac diagnosis, but raw recordings are often affected by movement artifacts, electrode misalignment, and electrical noise.

Research Gap. Traditional ECG report generation methods primarily rely on 1-D signal features, overlooking the rich visual information in ECG images that clinicians use for interpretation.

Poster Message. We use the ECG waveform, ECG image, and clinical text together, then align them through a trainable multimodal attention module.

2. Contributions

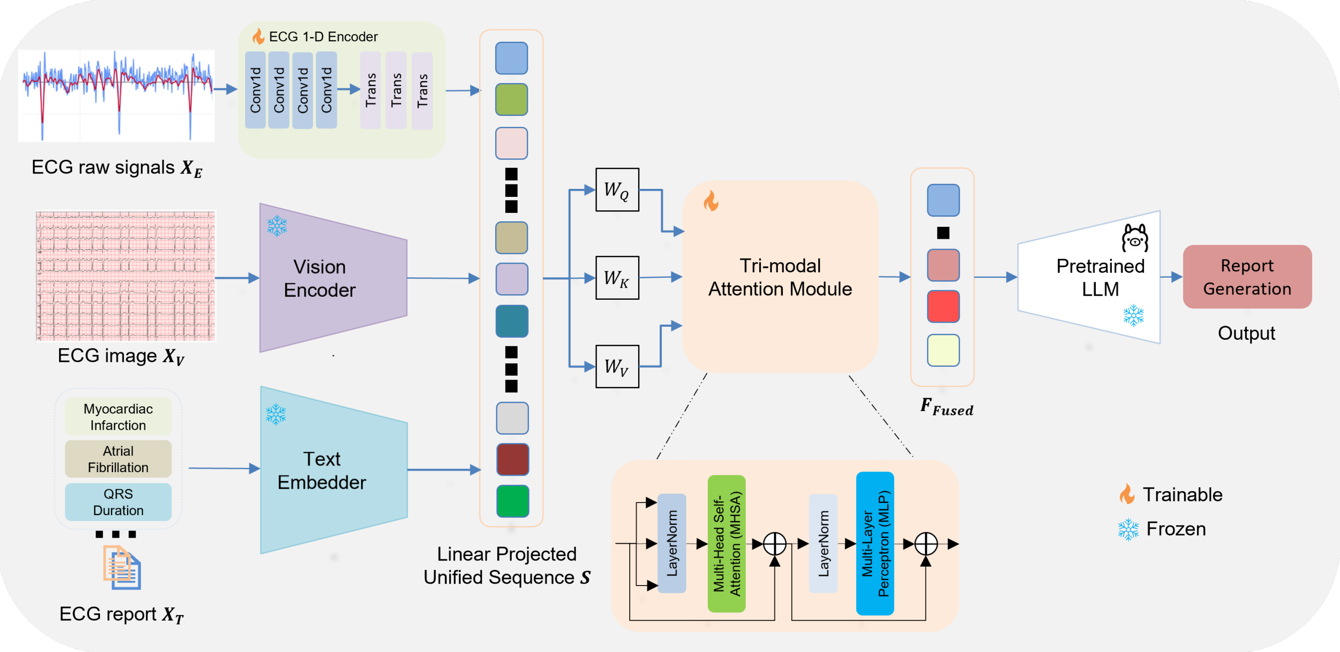
- A tri-modal framework integrating raw 12-lead ECG signals, visualized ECG images, and textual clinical reports.
- A unified feature sequence that supports both intra-domain and cross-domain attention.
- A lightweight adaptation strategy that reuses pretrained vision-language and language models (LVLMs and LLMs) instead of training a large model from scratch.

3. Experiment Settings

Dataset	MIMIC-IV ECG with MEETI ECG images
Split	85% training, 15% evaluation
Metrics	BLEU-1/2/4 and ROUGE-L/1/2
Training	100 epochs with AdamW, cosine LR, early stopping
Hardware	Single NVIDIA RTX 5090 GPU

Compared configurations.
Vision encoders: LLaVA-Next, SigLIP, DINOv2-Giant, Biomed-CLIP.
LLM interpretation heads: LLaMA-3 and Mistral-Nemo.

4. Methodology



Multimodal Attention. Project each modality into one shared sequence, attend across modalities, then apply residual fusion.

$$\textbf{Align:}\quad S = \mathrm{Concat}(H_V, H_T, H_E), \qquad Q = SW_Q,\ K = SW_K,\ V = SW_V,$$

$$\textbf{Attend:}\quad Z_{\text{fused}} = \mathrm{Softmax}\left(\frac{QK^{\top}}{\sqrt{D_{\text{shared}}}}\right)V,$$

$$\textbf{Fuse:}\quad F_{\text{fused}} = \mathrm{LayerNorm}(S + Z_{\text{fused}}).$$

ECG Encoder
1-D CNN captures local morphology; Transformer blocks capture temporal dependencies.

Vision Encoder
Pretrained visual models encode grid patterns, waveform shapes, spikes, and visual structure.

LLM Task Head
Fused multimodal features are mapped into the LLM space for ECG report generation.

Design Rationale. The visual domain complements the raw signal and text domains, allowing the model to reuse pretrained VLM/LLM representations while learning a compact multimodal alignment layer.

5. Results

0.727 Best Mean Score

0.773 Best ROUGE-L

0.658 Best BLEU-4

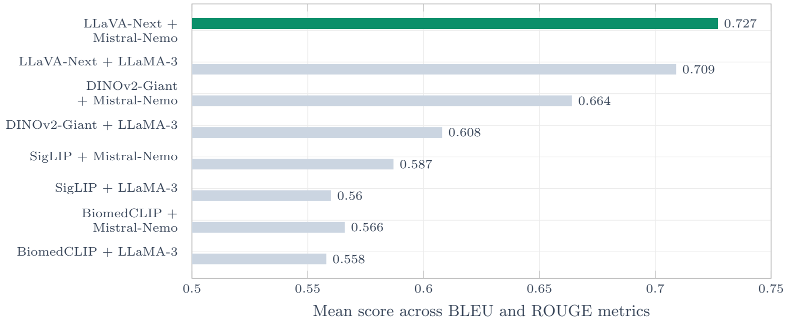


Table 1. Natural language generation results on MEETI.

VLM	LLM	B1	B2	B4	RL	R1	R2
LLaVA	LLaMA-3	*0.722*	*0.687*	*0.625*	*0.759*	*0.780*	*0.683*
LLaVA	Mistral	**0.745**	**0.701**	**0.658**	**0.773**	**0.786**	**0.696**
SigLIP	LLaMA-3	0.632	0.538	0.461	0.556	0.601	0.574
SigLIP	Mistral	0.648	0.572	0.491	0.583	0.628	0.602
DINOv2	LLaMA-3	0.651	0.582	0.502	0.637	0.673	0.601
DINOv2	Mistral	0.682	0.639	0.579	0.710	0.701	0.673
BioCLIP	LLaMA-3	0.629	0.535	0.479	0.548	0.593	0.565
BioCLIP	Mistral	0.636	0.535	0.449	0.573	0.612	0.593

Discussion and Future Works. LLaVA-Next + Mistral-Nemo is strongest overall, while BiomedCLIP remains competitive with SigLIP despite fewer parameters, emphasizing suitability for resource-constrained medical environments.
Future works will expand the framework to other tasks such as ECG signal quality assessment and ECG denoising with non-linear extracted features.

Key Takeaways: Aligning raw ECG signals, ECG images, and clinical reports enables stronger ECG report generation while reusing pretrained VLMs and LLMs with a compact fusion module.